Labor Market Value of College Networks*

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Abstract

Do college alumni networks within a firm confer labor market benefits? Using data from a large online job platform, we construct a panel data of workers' job histories and the size of undergraduate alumni networks at their firms. We assess whether having a larger alumni network at the firm impacts their future job outcomes with a saturated fixed effects model. We find that larger alumni networks increase an individual's likelihood of promotion, moving firms, and job title improvement in the 3 to 5 years after moving. Our analysis supports productivity gains and referrals as plausible mechanisms of these effects. Understanding college alumni networks as a dimension of the returns to college has important implications for human capital investment decisions, such as college enrollment and college choice.

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1 Introduction

A college education offers large benefits on the labor market, including earnings premium and higher likelihood of employment.[1] These effects vary both across colleges, by quality of institution, and by majors (Webber, 2014; Dillon and Smith, 2020; Altonji et al., 2023; Chetty et al., 2023). Although the literature documents heterogeneity in returns to college, most studies on this topic model labor market benefits of college as static; for example, as a fixed earnings premium. However, the college one attends can also have dynamic labor market returns throughout an individual's career.

In this paper, we study one such potential benefit: college alumni networks at firms as another dimension of returns to college that may yield dynamic labor market benefits. These benefits vary as individuals progress in their career over time, through promotions within firms and job mobility across firms. Within a firm, alumni networks can act as social networks, which have been shown to impact labor market outcomes (for example, see review article by Beaman (2016)). As individuals develop their careers and spend time at different firms, a connection with fellow alumni at each firm may change the pace of career progression due to productivity gains from peer effects, referrals for job opportunities, or by providing better information to firms about unobserved worker productivity. Therefore, examining the value of college alumni networks helps us understand an important dimension of the dynamic labor market benefits of college choice.

This paper studies the impact of college alumni networks on employees' career progression by utilizing data from a large online job platform with information on each worker's college, place of work, and their job title throughout their career. We construct a measure of undergraduate alumni networks and use a saturated fixed effects model to assess whether the size of an individual's alumni network at the current firm impacts her likelihood of promotion, job mobility, and, when moving jobs, whether she improves her job standing at a different firm. We then conduct heterogeneity analyses to assess mechanisms through which alumni networks affect these outcomes.

[1] Among young full-time workers (age 25 to 34) in 2022, the median worker with a Bachelor's degree or higher earned 70 percent more and were 6.7 percentage points more likely to be employed than the median worker with high school as the highest level of education. Source: Digest of Education Statistics, 2023

First, we show that college alumni networks, not just college education itself, play a role in labor market outcomes. Existing studies on networks in labor markets focus on social networks, defined by shared demographic characteristics, such as ethnicity, or connections through organizations, such as online social networks or previous firms (Hellerstein et al., 2011; Beaman, 2011; Beaman and Magruder, 2012; Dustmann et al., 2015; Hensvik and Skans, 2016; Armona, 2025). We find that a one standard deviation increase in alumni network size increases promotion by 5.9 percent and moving firms by 12.3 percent.

Then, we provide evidence that the labor market effects of college alumni networks go beyond the referral mechanism. Most research models the function of social networks in labor markets as a network of job referrals that increases job opportunities of connected people (Calvó-Armengol and Jackson, 2004) or reduces information frictions about job seekers for firms (Montgomery, 1991). However, alumni networks can boost productivity through opportunities for learning from peers or mentorship. In contrast to referral networks and relaxation of information constraints, productivity gains imply that benefits of alumni networks would persistent over the individual's career, even when she moves to a firm with a small network.

Indeed, our findings support productivity gains and referrals as plausible mechanisms for the labor market benefits of alumni networks. Specifically, college alumni networks from an individual's past firm continue to positively impact career growth at her new firm even after controlling for the new firm's alumni network. This finding suggests that individuals' receive portable productivity gains from networks that they can move across firms. We also find that individuals with larger senior networks earn promotions faster, but movers do not experience any benefits from larger senior networks. This suggests that the gains from a senior network are derived from within-firm referrals for promotions and are not portable across jobs, such as productivity gains. We do not find evidence to support networks relaxing information constraints about unobserved productivity or that employee preferences for networks drive the impacts on career outcomes.

Our unique panel data and saturated fixed effects model addresses sorting, the primary concern with empirical studies on the impact of networks. Sorting on unobserved characteristics to both colleges and firms will confound any estimated effects of alumni networks.

However, because of the rich job history information and the panel nature of our data, we are able to net out any static individual or dynamic firm productivity by including individual and firm-year fixed effects. In a world where higher ability individuals sort to "better" colleges and more productive firms, our estimation strategy would yield null effects if alumni effects are primarily driven by sorting. Intuitively, sorting would lead to smaller alumni networks of high-ability students at less productive firms. In these cases, we would not estimate any correlation between network size and employee outcomes. As we estimate positive effects of network size, we can reject sorting as the primary driver of the labor market benefits of alumni networks.

Our findings have important implications for college enrollment and college choice. While the benefits shared within a college network may be similar to other social networks, the cost of access to an alumni network are significantly higher due to the rising costs of college. The potential benefits of college alumni networks therefore may be an important parameter to understand for college enrollment and college choice decisions.

In addition, our paper also has important implications for both workers and firms. For example, our finding that college alumni networks, as a form of social network within the firm, confers long-term productivity benefits while improving referrals can inform firms and workers on how to value these networks. As a practical example, workers may face a trade-off between current wages and future wages when choosing between jobs at firms with smaller or larger alumni networks.

The rest of the paper proceeds as following. Section 2 summarizes the data and Section 3 outlines our identification strategy. Section 4 presents results for short-run career outcomes and Section 5 discusses mechanisms. We conclude in Section 6.

2 Data

We use data from Lightcast Profile Analytics, which provides employee profiles in the United States containing information similar to an online job platform. For this study, the most relevant variables include job histories with start and end dates, position titles, and education history. We also construct two derived variables central to our analysis, firm-specific college alumni network size and position seniority.

We construct a person-year panel dataset from people's job histories. To do so, for each person-year i, t, we take person i's firm of employment f in year t from their job history's start and end dates.[2] For people who switched jobs during a given year, the job they held at the end of the year is taken as their firm of employment j at year t. For people who held multiple firms in a given year, we keep only the firm they worked at for the longest duration. Thus, each person has one job observation in each year.

We restrict our sample to employees who worked at one of the largest 500 firms at least once in their employment history. Moreover, we include private sector jobs only and exclude firm types that have unique labor markets and hiring practices.[3] Our resulting sample consists of 6.94 million individuals with 63.3 million person-year observations from 2000 to 2024 for people who received their Bachelor's degree in 1980 or later. The average worker in the sample graduated from college in 2007, has spent 13.4 years in the labor market after college, and has worked at 2.11 firms between 2000 and 2024.

We now describe how we construct the two derived measures for our analysis: firm-college networks and seniority.

2.1 Firm-College Networks

For firm f in year t, we define the total number employees as N_{ft} and the number of employees that are undergraduate alumni from college j as N_{jft}. Thus, the share of employees in f, t that are alumni of j is defined as $S_{jft} \equiv \frac{N_{jft}}{N_{ft}}$. We similarly define the number of new hires in f, t (n_{ft}), new hires from college j (n_{jft}), and share of new hires from college j (s_{jft}).

To understand the measure better, we provide some descriptive facts about college networks at firms. As shown in Table 1, the average college network at a firm (N_{jft}) consists of 4.28 employees with average new hires from college (n_{jft}) being 0.66 employees. Networks are small relative to firm size but vary significantly, with the largest college-firm pair share (S_{jft}) representing 37 percent of the firm's employees.

[2]For job spells with missing dates, we impute them using available school and job start and end dates, (e.g., if the person has firm A's start date as 2004 and end date missing, and firm B's start date as 2006, then we impute that the person worked at firm A from 2004 to 2006). We exclude people from the sample whose missing information made imputation of dates unreliable.

[3]We exclude universities, hospitals, real estate agencies, multi-level-marketing, government, religious, and non-profit organizations. These job spells occur as "holes" in the person-year observations.

We also find that alumni networks are persistent. To examine this, we estimate the following equation that relates share of new hires to share of the college network stock,

$$s_{jft} = \rho S_{jft-1} + \mu_j + \mu_f + \mu_t + \epsilon_{jft} \tag{1}$$

where μ_j, μ_f, μ_t are college, firm, and time fixed effects, respectively. The coefficient ρ captures the correlation between a college's alumni network size, relative to the firm size, and the share of newly hired employees from that same college in the subsequent year.

Estimates reported in Table 2 show that the existing stock of employees from a specific college at a firm is highly correlated with the flow of new employees from that college. After including college, firm, and year fixed effects, a network that is larger by 10 percentage points of the firm's size is associated with the share of new employees from that college being 8.12 percentage points larger. This large correlation implies that a college alumni network size at firms may be a determinant of hiring from that college. However, it is not clear from the firm level analysis whether the correlation between existing alumni networks and new hiring is due to labor market benefits of networks or due to sorting of students to specific colleges and firms. In the following sections, we conduct individual level analyses to assess the causal effect of alumni networks on labor market outcomes.

2.2 Seniority as Measure of Career Progression

While Lightcast includes rich information on individuals' employment history, we do not observe wages. Instead, we leverage individuals' job titles to measure career progression. Specifically, we measure each job's seniority following the methodology in Amornsiripanitch et al. (2023) and Amornsiripanitch et al. (2025). Within an industry and firm size decile, seniority is defined as the number of post-college years it takes the median worker to attain a specific title. In the absence of wages, seniority provides information about employee's career progression that is necessary to assess the labor market impacts of college alumni networks. Specifically, the seniority measures allow us to compute whether the person's current job spell is an "improvement" (in the sense of career progression) compared to their previous spell. This is especially useful when the person changes their firm of employment or occupation, since title changes across firms and occupations are harder to interpret on their own.

Several stylized facts from Amornsiripanitch et al. (2023) and Amornsiripanitch et al. (2025) reassure us that seniority is a reasonable measure of labor market productivity. First, seniority is positively correlated with estimated wage. Second, individuals who attended elite colleges have faster career progression, as measured by seniority, than individuals who attended lower ranking colleges. Third, receiving an MBA degree, especially from elite institutions, allows individuals to obtain higher seniority jobs after graduation. Fourth, holding fixed individuals current estimated wages, individuals who hold higher-seniority jobs are able to transition into new jobs that have higher wages. Lastly, the distribution of seniority values in the Lightcast data set mirrors the idea of a corporate pyramid; that is, most individuals' careers stop progressing after they have reached seniority levels of around 12 years, which is consistent with the idea that many people get stuck in "middle management," while only a few exceptional individuals achieve jobs with very high seniority levels (e.g., more than 20) such as distinguished scientist or executive chairman.

Table 3, Panel A shows that the average worker is employed in a position with seniority 9.08; that is, it takes a little over 9 years for the median worker in the same industry and firm size decile to reach the worker's position.[4] This is consistent with our average labor market experience also being around 10 years. Using this seniority measure, we additionally define two variables: *Promote*, which identifies whether an employee's job title changed and their seniority measure increased within their firm, and *Improve*, which, for those who move firms, measures whether a worker's seniority measure increased when moving (i.e., vertical move).

3 Empirical Model

To assess the impacts of college alumni networks on career progression, we estimate Equation 2. For individual i who graduated from college j observed working for firm f at time t,

$$Y_{ijt+1} = \beta \ln N_{jft} + \gamma X_{it} + \mu_i + \mu_{ft} + \epsilon_{ijt} \tag{2}$$

College network size, N_{jft}, is defined as before, measuring the number of employees from the individual's college j, including herself, working at her firm f in time t. The regression also includes controls, X_{it}: a quadratic in potential experience and two-year bins of the

[4] A negative seniority occurs when the median worker reaches that position with less than 4 years of college.

employee's tenure at the firm. The terms μ_i and μ_{ft} are individual and firm-year fixed effects, respectively. Standard errors are clustered by firm.

Outcomes Y_{ijt+1} include indicators for whether the individual was promoted at the same firm ($Promote_{it+1}$), moved to a different firm ($Move_{it+1}$), or improved their job conditional on moving to a different firm in the subsequent year ($Improve_{it+1}$). As described above, promotion is defined as changing job titles within the same firm to a position with higher seniority. If an individual switches firms, we define the individual as having improved their position if the new position has higher seniority than the position at the previous firm. As shown in Table 3, Panel A, 4.6 percent of individuals earn promotions annually, 9.8 percent move to a different firm, and 52.8 percent of firm movers improve to a more senior position.

The coefficient of interest β measures the impact of the current size of worker's college networks on employment outcomes in the next year. Identification of β as the causal effect of networks relies on the fixed effects capturing unobserved factors that may determine employment outcomes and are correlated with network size. Selection on individual unobserved ability is a primary concern. Impacts of networks on outcomes may be driven by sorting of high ability students to similar colleges and firms. We address for this identification concern with the individual fixed effect μ_i, which accounts for any permanent unobserved ability or productivity. The firm-year fixed effect μ_{jt} accounts for the concern that firms' demand for labor may confound the network effect. For example, employment opportunities may vary based on years when firms are more active in hiring or based on firms that have consistent recruitment relationships with specific colleges. Intuitively, the inclusion of both fixed effects means that the variation identifying β results from differential changes to individuals' college network size relative to other employees' college network sizes within the same firm-year.

Importantly, this empirical specification allows us to separate the impact of alumni networks from sorting. A correlation between large networks and career outcomes, as in Equation 1, may not sufficiently identify the impact of networks on career progression if students sort to similar colleges and firm based on unobserved characteristics that simultaneously improve their education outcomes and labor market productivity. Such sorting may lead to the results observed in Table 2, where larger networks within firms tend to hire more from those colleges.

However, if labor market gains from networks are entirely due to sorting, we should expect β in Equation 2 to be zero.

In a simple framework, consider a labor market with two colleges (l and h) and two firms (L and H). High ability students sort to college h and firm H, while low ability students sort to college l and firm L. As a result college h network size is large at firm H and college l network size is large at firm L. If all the effects are driven by sorting, employees from college h should have better outcomes at both firms H and L, and thus β would be uncorrelated with alumni network size. However, if alumni networks improve labor market outcomes, college l alumni at firm L would have better outcomes than college h alumni at firm L because the l network is larger. As a result, in a within-firm analysis such as Equation 2, sorting as the primary determinant of outcomes would yield a β of zero while alumni networks presenting labor market benefits would produce a positive estimate of β.

While this identification strategy addresses the primary confounding factors discussed above, it is unable to account for time varying unobserved heterogeneity that may be correlated with college networks. Differential learning behaviors correlated with network size would be an example of such heterogeneity. Therefore, students from colleges with large networks may progress faster in their careers if they learn faster on the job than students from small networks. Although this is a limitation of our strategy, note that the identification threat presented is from the time varying portion of this heterogeneity and not from permanent heterogeneity, such as individuals who are "fast" learners throughout their career, that may drive differential learning due to the inclusion of individual fixed effects.

4 Labor Market Effects

Table 4 reports estimates of β that measures the impact of college alumni network size on the following year's labor market outcome for our three outcomes. Our results show that college networks have a positive and statistically significant effect on the likelihood of promotion and switching firms but a statistically and economically insignificant effect on the likelihood of improvement in the next year for those switching firms.

To interpret the marginal effects, consider the following. For the average employee, who has 65 college alumni at their firm, adding one additional person to the network only increases

the likelihood of promotion by 0.003 percentage points and the likelihood of switching firms by 0.01 percentage points.[5] The implied elasticity of promotion and moving firms with respect to college network size is 0.043 and 0.092, respectively.[6]

While the marginal effect of adding one employee to a college network are economically small, considering the high variance in network size and the compounding effects over the employee's duration at the firm shows that the results are economically meaningful. First, college alumni network size at firms vary greatly—the standard deviation is 185. A one standard deviation change in network size for the average network size of 65 increases the annual likelihood of promotion by 0.27 percentage points and annual likelihood of switching firms by 1.21 percentage points. Relative to the average promotion and moving rates, this one standard deviation change in network size increases promotion by 5.9 percent and moving by 12.3 percent. As a reference, moving one standard deviation in the distribution of college value-added increases wages by 3 percent (Mountjoy and Hickman, 2021). Second, the average employee spends 6 years at a firm with an annual promotion rate of 4.6 percent, which implies a 24.6 percent likelihood of earning any promotion during those six years.[7] A one standard deviation difference in network size increase the rate of earning any promotion during the 6 years by 1.3 percentage points.

5 Mechanisms

Larger alumni networks may result in labor market benefits for various reasons. First, alumni networks may boost employee productivity, both permanently and during their time at the firm. For example, closer relationships to colleagues due to a common alma mater can facilitate learning on the job that develops human capital. These relationships can also help identify complementarities in production and allocate employees efficiently to firm projects.

Second, alumni networks may provide information about unobserved productivity. As shown in our descriptive results, firms tend to hire from colleges that have larger internal

[5] Adding one employee to a network of 65 employees leads to a $\ln(66) - \ln(65) = 0.015$ log point change in network size. Multiplying the 0.002 and 0.009 coefficients for promotion and moving, respectively, results in a 0.003 percentage point and 0.01 percentage point change. In relation to the average promotion rate of 4.6 percent and moving rate of 9.8 percent, a one person addition to the network increase promotion and moving likelihood by 0.07 and 0.14 percent, respectively.

[6] Elasticities are calculated as $\frac{\beta}{\text{Outcome Mean}}$.

[7] Over six years, the rate of earning any promotion is $1 - (1 - 0.046)^6 = 0.246$.

alumni networks. Existing networks within a firm provide incentives to continue hiring from those colleges as better information about the unobserved productivity of job applicants reduces uncertainty in the potentially risky investment of hiring. Third, and relatedly, larger networks may provide more referrals, which may increase the choice set of employees searching for jobs as well as increase competition for the same jobs when network sizes are large.

Last, employees may hold preferences for working with peers from their college. While we cannot directly decompose which of these mechanisms is most important, in the following sections we conduct empirical tests for the evidence of some of these mechanisms.

5.1 Long-Term Productivity Effects

College networks may provide both long-term and match-specific productivity benefits. If college alumni networks only improve employees' match-specific productivity, then these effects should subside when workers leave the firm. In contrast, if there is an effect to overall productivity (not just match-specific productivity), we would expect to observe a long-term and persistent effect on labor market outcomes even after they change firms.

In Table 4 column 1, we saw evidence, through higher rates of promotions, that match-specific productivity effects may exist. Yet, there may be additional long-term benefits as well. For this reason, we focus on job improvement for movers as the outcome of interest in this section. We investigate the effect of alumni network size at the job mover's origin firm in year t on long-term improvements 3 years $(t+3)$ and 5 years $(t+5)$ post-move. We construct the 3- and 5-year improvements relative to both the job at the original firm in year t and the first year of the job at the new firm $(t+1)$ for the moving employee.

Table 5 shows that the network size in the original firm does improve job seniority for movers, 3 and 5 years after the move. Specifically, column 4 shows that network size has a positive and statistically significant increase on the likelihood of holding a more senior job relative to the job at the origin firm (t) in the five years after moving firms. The economic magnitude is small, as the implied elasticity of improvement in $t+5$ is 0.003. However, the impact on improving jobs relative to the first year at the new firm $(t+1)$ is more economically meaningful. The elasticity of improvement in $t+3$ and $t+5$ relative to $t+1$ is 0.029 and 0.015, respectively. Combined with results from Table 4 column 3, the result suggests that larger networks may lead to an initial lateral move (no improvement from t to $t+1$), but eventually

leads workers to progress faster in their newer jobs. Therefore, this finding provides support for college alumni networks yielding long-term employee productivity gains.

However, long-term gains from the networks at the original firm may be driven by correlation with the network at the new firm. We consider this channel by controlling for network size at both the origin and destination firm in Table 6. Conditional on the network at the new firm, the network size at the old firm still has a positive impact on likelihood of improving in $t+3$ and $t+5$ relative to the first year at the new firm. The results hold with a more stringent set of fixed effects, in columns 3 and 4, that include origin-by-destination firm fixed effects. This specification estimates the impact of alumni network size on improvement for job movers who are making the same origin firm-to-destination firm move. The 3- and 5-year elasticities for improvement for the origin firm's network size is 0.025 and 0.013, respectively. Therefore, alumni networks from old firms continue to matter for workers' career growth at new firms even after conditioning for the network at the new firm and after holding the firm-to-firm move fixed. These findings support that networks provide long-term productivity gains. They also support the fact that the effect on promotion (Table 4 column 1) may not only be driven by match-specific productivity gains.

5.2 Networks as Information On Workers

A potential explanation for the positive effects on job promotion and mobility is that networks provide advantage to connected workers by allowing employers to better observe their productivity (information channel). To understand whether alumni networks operate purely as information about employee's unobserved productivity, we estimate the labor market effects of college networks over the life cycle of an individual's career using a test similar to Altonji and Pierret (2001). If networks offer benefits solely due to information, the impact of alumni networks on labor market outcome should diminish over workers' career as firms receive more information about their unobserved productivity from their job histories (e.g. resumes, internal evaluations, personal recommendations) unobserved to the econometrician.

Specifically, we modify Equation 2 to include interaction terms of alumni network size with the employee's experience, measured in 2-year bins,

$$Y_{ijt+1} = \sum_k \beta_k \times \delta_{itk} \times \ln N_{jft} + \beta \ln N_{jft} + \gamma X_{it} + \mu_i + \mu_{ft} + \epsilon_{ijt} \tag{3}$$

where $\delta_{itk} = 1$ if i's experience is in two-year experience bin k at time t, and 0 otherwise. The coefficients β_k are the effect of alumni network for workers in experience bin k.

Figure 1 plots our estimated β_k. We do not find evidence to support the information channel as the only explanation of the benefits of alumni networks. Depending on the outcome, network effects even seem to strengthen over the life cycle. The positive effect on promotions does not appear until after 2 years of experience, but thereafter the effect size remains similar across the life-cycle. The slightly larger increase in promotion reflects higher promotion rates early in the career (Table 9). After 2 years of experience, the elasticity of promotion ranges from 0.047 early in the career to 0.077 after 20 years of experience.

The life-cycle gradient in the impact of alumni networks on job switching is much more pronounced with elasticities increasing from 0.093 in the first two years of experience to 0.156 after 20 years of experience. Similarly, network size has *negative* effects on likelihood of job improvement in the next year for movers with less than 6 years of experience while having no effect on likelihood of improvement for later career movers. A similar directional pattern exists for long-term improvement but the gradient is more muted.

5.3 Referrals

The role of networks in labor markets is frequently thought of as providing referrals for job opportunities (Montgomery, 1991; Calvó-Armengol and Jackson, 2004; Dustmann et al., 2015). Specifically, the referral network model by Calvó-Armengol and Jackson (2004) implies that there are two competing effects of social networks: while larger networks provide more referrals, they also create more competition for those jobs.

We test for this mechanism by leveraging our unique data to create two separate college alumni network measures. Specifically, we split our network measure N_{jft} into two, one calculated using only workers that is more senior than her (N^+_{ijft}) and one calculated using

only workers at or below the same seniority level as her (N^{-}_{ijft}). Then we estimate the impact of the two alumni network measures on job outcomes.

$$Y_{ijt+1} = \beta^{+} \ln N^{+}_{ijft} + \beta^{-} \ln N^{-}_{ijft} + \gamma X_{it} + \mu_i + \mu_{ft} + \epsilon_{ijt} \tag{4}$$

Table 7 presents the results for our three outcomes of promotion, move, and improvement indicators. Column 1 shows that more senior networks increase the likelihood of promotion. The economic magnitude is also much larger than previous estimates of the effect of overall network size – the elasticity of promotion with respect to the size of the senior alumni network is 0.289. Conversely, larger networks of less senior alumni reduces the likelihood of promotion, potentially driven by higher competition for limited roles.

These results are exactly what Calvó-Armengol and Jackson (2004) would predict. However, the large positive effect of senior networks can also be due to both productivity and referrals. Senior alumni may provide mentorship that improves human capital and also be better connected to provide referrals for career growth opportunities within the firm. However, if this senior network effect was driven by productivity gains, we would expect similar effects for the likelihood of moving and improvement. Instead, we observe the opposite. Senior networks reduce the likelihood of moving and have no effect on job improvement in columns 2 and 3, respectively. Taken together, the findings do not support that senior networks improve productivity through mentorship. Therefore, we conclude that alumni networks do provide labor market benefits through referrals.

5.4 Worker Preferences

Last, the estimated network effects may be driven by worker preferences. If employees derive utility from working with peers from their colleges (e.g., homophily), we might expect them to continue progressing to firms with larger networks. However, as shown in Table 4, employees are more likely to move to a different firm when their networks are large, and when they move, they are unlikely to make vertical moves (at least initially). If workers had preferences for larger networks, workers would move only when they are being compensated for the loss of utility.

In order to more directly see whether workers progress to firms with larger networks, we estimate the impact of past network size on future network size for job movers. Results in Table 8 show that employees move to firms with *smaller* alumni networks. The finding is robust to choice of outcome as absolute network size, network size relative to firm size, or an indicator of moving to a larger network. All of these results are opposite of what we might expect under strong employee preferences for alumni networks; therefore, we conclude that preferences are not driving the labor market results.

6 Conclusion

This paper studies the impact of college alumni networks on employees' career progression. We find that alumni networks are persistent within firms and positively impact the employee's likelihood of promotion, movement across firms, and long-term job improvement. Our analysis supports productivity gains and referrals as plausible mechanisms for the labor market benefits of alumni networks. College alumni network benefits from an individual's past firms persist even at new firms, and larger senior networks increases the likelihood of within-firm promotions. Findings do not support networks relaxing information constraints about unobserved productivity or that employee preferences for networks as determinants of career outcomes.

This research has important implications as it helps us understand an additional dimension of the returns to college. In contrast to traditional estimates of the returns to college, the labor market benefits of alumni networks are dynamic as employees move across firms and networks change over time. Measuring these benefits provide information for costly human capital investment decisions, such as college enrollment and college choice. Our result that the size of the network matters implies that attending larger and cheaper public universities may provide more net benefits than smaller but more expensive private universities.

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A Tables

Table 1: Firm-School Dataset Summary Statistics

Variable	Obs	Mean	Std. dev.	Min	Max
Firm New Hires from College j	9,157,620	0.66	3.23	0	806
Firm New Hires	9,157,620	1,827	2,927	0	42,337
Share of Firm New Hires from College j	9,157,620	0.001	0.003	0	1
Firm Employees from College j, t-1	9,157,620	4.28	20.5	0	4,212
Firm Employees, t-1	9,157,620	11,763	14,898	27	173,049
Share of Firm Employees from College j, t-1	9,157,620	0.001	0.002	0	0.369

The sample only includes the largest 500 firms, 2015 - 2024.

Table 2: Firm-School Regression

	(1)	(2)	(3)	(4)
	Share of New Hires This Year			
Last Year's Share of Employees	0.841***	0.839***	0.839***	0.812***
	(0.008)	(0.008)	(0.008)	(0.009)
Observations	9,157,620	9,157,620	9,157,620	9,157,620
R-squared	0.449	0.449	0.449	0.454
Firm FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
College FE	No	No	No	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The regression does not include any unlisted control variables. Standard errors in parentheses are clustered by company.

Table 3: Individual-Year Dataset Summary Statistics

Panel A: Person-Year	Obs	Mean	Std. dev.	Min	Max
Promotion	56,722,217	0.046	0.209	0	1
Move	63,316,339	0.098	0.298	0	1
Improve	3,445,998	0.528	0.499	0	1
Experience	63,316,339	9.99	6.08	2	44
Seniority	63,316,339	9.08	6.48	-4	36.5
Tenure in Position	63,316,339	3.99	3.43	1	20
Network Size	63,316,339	64.8	185	1	3870
Network Size - More Senior	63,316,339	15.5	56.3	0	2895
Network Size - Same or Less Senior	63,316,339	36.4	110	1	2982

Panel B: Person	Obs	Mean	Std. dev.	Min	Max
Graduation yr (everyone)	6,699,029	2007	10.0	1981	2024
Graduation yr (working sample)	6,082,892	2007	9.94	1981	2023
Years in labor market (observed in data)	6,937,025	13.4	6.76	3	44
Number of firms	6,937,025	2.11	1.15	1	12
Number of company-	6,937,025	2.63	1.48	1	16
Average duration in firm	6,937,025	6.24	4.37	1	20

The sample includes person-year observations for people who ever worked at one of the largest 500 firms, 2000 - 2024. The sample size for graduation year is smaller since having that information was not a requirement for inclusion in the sample.

Table 4: Individual Promotion, Move, Improve

	(1) Promotion	(2) Move	(3) Improve
Ln(Stock of Employees Last Year)	0.002***	0.009***	-0.0006
	(0.000)	(0.000)	(0.000)
Outcome Mean	0.046	0.098	0.528
Observations	56,722,217	63,316,339	3,445,998
R-squared	0.224	0.398	0.461

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include controls for experience, experience squared, and company duration. All regressions include individual and firm-year (lagged) fixed effects. The sample in Column 3 only includes movers. Standard errors in parentheses are clustered by company.

Table 5: Long Term Improve Outcomes

	(1) Improve $t+3$	(2) Improve $t+5$	(3) Improve $t+3$	(4) Improve $t+5$
Ln(Stock of Employees Last Year)	0.007***	0.006***	0.0003	0.002***
	(0.001)	(0.001)	(0.001)	(0.001)
Outcome Mean	0.245	0.392	0.58	0.627
Observations	2,198,006	1,381,495	2,198,006	1,381,495
R-squared	0.516	0.558	0.562	0.629
Relative to period	$t+1$	$t+1$	t	t

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include controls for experience, experience squared and company duration. All regressions include individual and firm-year (lagged) fixed effects. The sample only includes movers. Improve t+3(5) conditions on a person having observations in periods $t, t+1, t+2$, and $t+3$ (and $t+4, t+5$). Ln(Stock of Employees Last Year) refers to period t. Standard errors in parentheses are clustered by company.

Table 6: Persistence of Networks

	(1) Improve t+3	(2) Improve t+5	(3) Improve t+3	(4) Improve t+5
Ln(Stock of Employees Last Year)	0.005***	0.004***	0.006***	0.005***
	(0.001)	(0.001)	(0.001)	(0.002)
Ln(Stock of Employees This Year)	0.001**	0.002***	0.003**	0.004**
	(0.001)	(0.001)	(0.001)	(0.002)
Outcome Mean	0.244	0.393	0.237	0.390
Observations	2,456,133	1,541,600	931,957	580,000
R-squared	0.504	0.554	0.693	0.718
Fixed Effects	Origin and Destination Firm		Origin-Destination Firm	

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include controls for experience, experience squared, and company duration. All regressions include individual and year (lagged) fixed effects. Outcomes are relative to period t+1. All regressions include seniority lag controls. Standard errors in parentheses are clustered by company.

Table 7: Relative Networks

	(1) Promotion	(2) Move	(3) Improve
Ln(Stock of Same or Lower Seniority	-0.009***	0.014***	0.001
Employees Last Year)	(0.000)	(0.001)	(0.002)
Ln(Stock of Higher Seniority	0.013***	-0.004***	-0.001
Employees Last Year)	(0.000)	(0.001)	(0.002)
Outcome Mean	0.045	0.100	0.536
Observations	56,722,217	63,316,339	3,445,998
R-squared	0.248	0.401	0.625

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include controls for experience, experience squared, and company duration. All regressions include individual and firm-year (lagged) fixed effects. The sample in Column 3 only includes movers. Standard errors in parentheses are clustered by company.

Table 8: Network Size as Outcome

	(1) Ln(Stock This Year)	(2) Share This Year	(3) Larger	(4) Larger
Ln(Stock of Employees Last Year)	-0.268***		-0.169***	
	(0.004)		(0.001)	
Share of Employees Last Year		-0.262***		-2.158***
		(0.008)		(0.031)
Observations	3,445,998	3,445,998	3,445,998	3,445,998
R-squared	0.682	0.616	0.635	0.578

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include controls for experience, experience squared, and company duration. All regressions include individual and firm-year (lagged) fixed effects. The sample includes movers. The "Larger" outcome indicates a strictly larger alumni network size (based on the stock of employees from the same school in column 3, based on the share of employees from the same school in column 4). Standard errors in parentheses are clustered by company.

Table 9: Outcome Means by Experience Bin

Experience Bin	Promotion	Move	Improve
1	0.02	0.03	0.64
2	0.05	0.13	0.56
3	0.06	0.13	0.55
4	0.05	0.12	0.53
5	0.05	0.10	0.53
6	0.05	0.09	0.52
7	0.04	0.09	0.51
8	0.04	0.08	0.50
9	0.04	0.08	0.50
10	0.04	0.08	0.49
11	0.03	0.08	0.48
12	0.03	0.06	0.47

Experience bins include two values up to bin 11, which includes values 21 - 25, and bin 12, which includes values above 25.

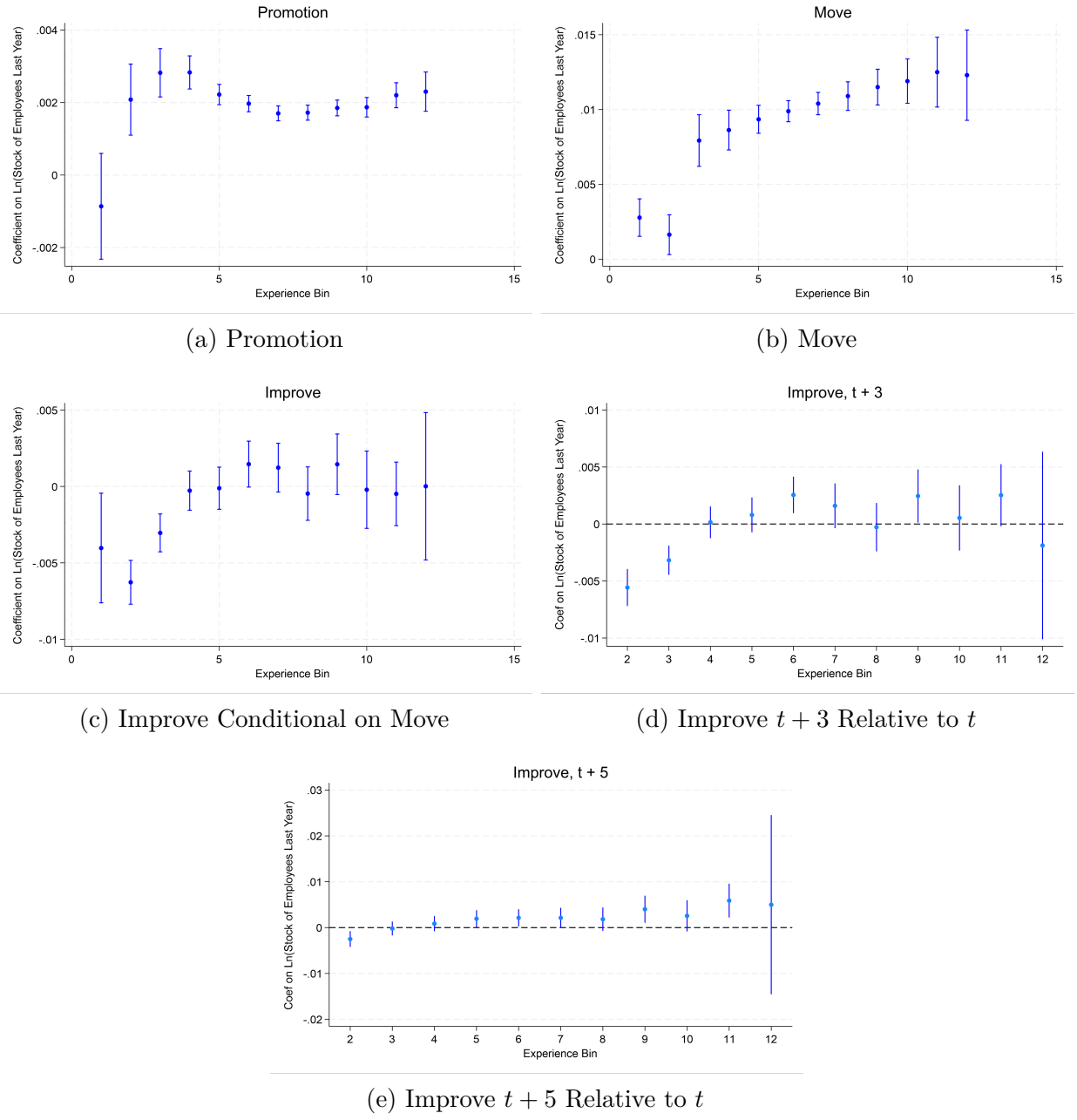
Table 10: Long Term Improve with Relative Networks

	(1) Improve t+5	(2) Improve t+3	(3) Improve t+5	(4) Improve t+3
Ln(Stock of Same or Lower Seniority	0.003**	0.006***	0.002*	0.000
Employees Last Year)	(0.002)	(0.001)	(0.001)	(0.001)
Ln(Stock of Higher Seniority	0.004**	0.002*	0.002*	0.000
Employees Last Year)	(0.002)	(0.001)	(0.001)	(0.001)
Outcome Mean	0.627	0.58	0.627	0.58
Observations	1,381,495	2,198,006	1,381,495	2,198,006
R-squared	0.56	0.516	0.741	0.699
Rel to pd	t + 1	t + 1	t	t

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include controls for experience, experience squared and company duration. All regressions include individual and firm-year (lagged) fixed effects. The sample only includes movers. Improve t+3(5) conditions on a person having observations in periods $t, t+1, t+2$, and $t+3$ (and $t+4, t+5$). Ln(Stock of Employees Last Year) refers to period t. Standard errors in parentheses are clustered by company.

B Figures

Figure 1: Life Cycle Graphs



(a) Promotion

(b) Move

(c) Improve Conditional on Move

(d) Improve $t+3$ Relative to t

(e) Improve $t+5$ Relative to t

Coefficients with 95% confidence intervals plotted. All regressions include lagged seniority and experience fixed effects (binned).